

Thermodynamic characterization of the boundary layer under fog and dew events in the coastal hyper-arid climate of the Atacama Desert

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2. Methods

2.1 Study Area and Data Acquisition

The Coastal Cordillera is a mountain range adjacent to the Pacific Ocean that spans most of the Chilean territory from north to south, reaching altitudes of approximately 1500 m above sea level (a.s.l.) (Lobos-Roco et al. 2024). Within this geographic context, the study area is located at Cerro Oyarbide (OYA; 20°29'0" S – 20°48' S), situated on the Coastal Cordillera above the Iquique International Airport, approximately 11 km inland from the coastline. At this site, hyper-arid conditions prevail, allowing *Tillandsia landbeckii* loma fields to develop across an undulated topography ranging between 900 and 1300 m a.s.l.) (Pinto, Barria, and Marquet 2006; García et al. 2021).

To capture the atmospheric variability along this complex terrain, an altitudinal transect consisting of eight meteorological stations was analyzed, as illustrated in Figure 1. These stations include OYA518, OYA780, OYA862, OYA1069, OYA1128, OYA1193, OYA1211, and OYA1354, where the numerical suffix explicitly indicates each station's operational altitude in meters above sea level. To establish a sea-level baseline for the Marine Boundary Layer (MBL) analysis, meteorological data from the Diego Aracena International Airport in Iquique (56 m a.s.l.; 20° 32' 28" S, 70° 10' 38" W) was incorporated (García et al. 2021). This coastal station is situated directly at the base of Cerro Oyarbide, sharing its latitudinal alignment.



Figure 1: Fotorrealistic 3D perspective of the altitudinal monitoring network across Cerro Oyarbide, showing the coastal baseline at Diego Aracena International Airport in Chucumata (56 m a.s.l.) and the stations scaling up the Coastal Cordillera.

All data management, statistical calculations, and subsequent environmental analyses in this investigation were executed using the R programming language, relying heavily on the `{tidyverse}` core ecosystem for data workflow optimization and synchronization. For this investigation, a continuous, full-year dataset for 2024 was utilized, with all surface environmental variables synchronized at a 10-minute temporal resolution. The data availability across the monitoring network varies by station and sensor type, as detailed in Table 1. Stations OYA518, OYA780, OYA1128, and OYA1211 exhibited excellent data continuity throughout 2024, maintaining availability close to or at 100% for baseline meteorological variables (air temperature, relative humidity, and wind vectors). Conversely, station OYA1069 presented critical data gaps due to sensor malfunctions, yielding an operational availability of only 16.7% for relative humidity and wind direction, and a total loss of temperature records.

Station OYA1211 serves as the central core node of the altitudinal transect. Positioned at the crest of a *Tillandsia landbeckii* field to characterize direct atmospheric-biospheric interactions

(García et al. 2021), OYA1211 features the most comprehensive instrumentation suite. This station provides specialized metrics for fog and dew analysis, including continuous visibility measurements (93.2% availability) and automated dew collectors, alongside complete vertical profiles for soil temperature and humidity, global radiation, and atmospheric pressure.

To complement these ground-based measurements with regional-scale observations, satellite-derived data was integrated to track the macro-scale presence of coastal fog and low clouds (FLC). Google Earth Engine (GEE) was used to process GOES-16 imagery covering the entire year of 2024, matching the 10-minute temporal resolution of the weather stations with a spatial resolution of 2 km (Lobos-Roco et al. 2024). FLC detection was executed through a cloud-computing application developed by the Centro UC del Desierto de Atacama, based on the dual-bandwidth thresholds proposed by Río et al. (2021) and Lobos-Roco et al. (2024). For diurnal imagery, the method combines visible and thermal bands to isolate cloudy pixels and distinguish low clouds from high clouds using a 273 K threshold. For nocturnal imagery, a brightness temperature difference (BTD) between the 3.8 μm and 10.9 μm infrared channels is applied with a -2 K threshold to differentiate low clouds from bare desert soil. This algorithm yields a binary output (1 for FLC presence, 0 for absence), providing a continuous spatial validation for the fog events recorded along the altitudinal transect.

Table 1: Data availability percentage (%) for the year 2024 at a 10-minute temporal resolution across the ground-based monitoring network. Note: The numerical suffix in each station’s name designates its operating altitude (m a.s.l.). Abbreviations: Temp (Air Temperature), RH (Relative Humidity), WS (Wind Speed), WD (Wind Direction), Fog (Ground fog event validation), Pres (Atmospheric Pressure), Rad (Global Radiation), Vis (Visibility), Dew (Dew collection). Dashes (-) indicate that the variable is not monitored by the station. Regional GOES-16 satellite imagery maintained a 100% data availability for Fog and Low Cloud (FLC) detection throughout the study period.

Station	Temp (°C)	RH (%)	WS (m/s)	WD (°)	Fog (%)	Pres (hPa)	Rad (W/m ²)	Vis (m)	Dew (mm)
AEROP.	91.5%	91.5%	14.5%	0.0%	-	91.5%	0.0%	0.0%	-
OYA518	99.4%	99.4%	97.4%	99.4%	99.4%	99.4%	99.4%	0.0%	-
OYA780	99.4%	99.4%	98.4%	99.4%	99.4%	99.4%	0.0%	0.0%	-
OYA862	86.0%	90.9%	86.1%	90.9%	8.5%	0.0%	0.0%	0.0%	-
OYA1069	0.0%	16.7%	0.0%	16.7%	16.7%	0.0%	0.0%	0.0%	-
OYA1128	100%	100%	100%	100%	100%	0.0%	0.0%	0.0%	-
OYA1193	90.2%	91.8%	90.2%	91.8%	91.8%	0.0%	0.0%	0.0%	-
OYA1211	100%	100%	100%	100%	100%	100%	100%	93.2%	100%
OYA1354	82.8%	83.3%	82.8%	83.3%	91.8%	0.0%	0.0%	0.0%	-

2.2 Classification of Fog and Dew Events

Distinguishing the exclusive atmospheric source of non-rainfall water inputs remains a significant challenge in arid environments, primarily because fog and dew events frequently overlap within the same diurnal cycles, and both standard fog collectors (SFC) and standard dew collectors (SDC) can simultaneously record minor water amounts (Lobos-Roco et al. 2024). To overcome this limitation and decouple both phenomena, this study implements a robust reclassification framework for water collection events by integrating high-resolution ground measurements and remote sensing observations through a hierarchical decision tree.

The physical distribution of the monitoring core node at the OYA1211 station highlights the proximity and synchronization between the instruments, featuring the standard dew collector (SDC), the standard fog collector (SFC), and the horizontal visibility sensor (VIS) operating simultaneously within the same local atmospheric microenvironment (Figure 2).

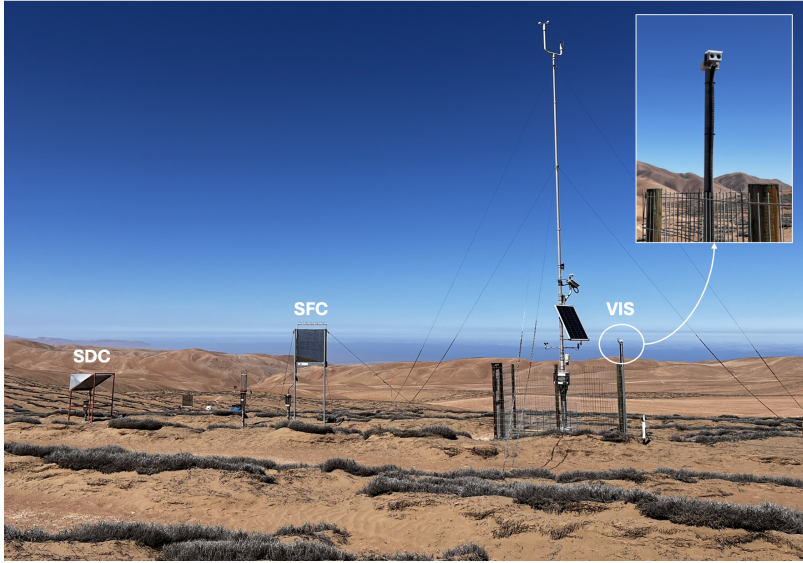


Figure 2: Field instrumentation setup at the OYA1211 monitoring station, showcasing the standard dew collector (SDC), standard fog collector (SFC), and the horizontal visibility sensor (VIS).

The classification algorithm considers every timestamp with active water collection in either the SFC, the SDC, or both instruments. To determine whether the collected water originated from a fog or dew event, horizontal visibility and satellite-derived FLC data are utilized as independent environmental tracers, prioritizing ground-based visibility data due to its high local precision.

To optimize target detection, the visibility sensor is installed perpendicular to the predominant advection path of coastal fog, which moves from West to East across the Coastal Cordillera;

consequently, the sensor’s optical axis is oriented along a North-South trajectory (Figure 2). Following international meteorological standards and fog-dynamics literature, a strict threshold of 1000 m is established to define fog conditions (Smith et al. 2022). Accordingly, any collection event coinciding with a horizontal visibility below 1000 m is strictly classified as a fog event, whereas events with visibility exceeding 1000 m are categorized as dew events, following the mathematical criteria described in Equation 1:

$$\text{Event Type} = \begin{cases} \text{Fog,} & \text{if Visibility} < 1000 \text{ m} \\ \text{Dew,} & \text{if Visibility} \geq 1000 \text{ m} \end{cases} \quad (1)$$

In instances where visibility data is unavailable due to sensor maintenance or data gaps, the satellite FLC binary indicator from GOES-16 is used as a secondary classification layer. Although GOES-16 provides continuous regional coverage, it is treated as a secondary preference due to its coarser spatial resolution (2 km) and inherent vertical uncertainty, as a detected low cloud deck may not necessarily intercept the complex surface topography where the collectors are situated (Lobos-Roco et al. 2024). Under this secondary branch, if water collection occurs in the absence of visibility data, the event is classified as fog if the GOES-16 pixel confirms the presence of a low cloud (FLC = 1). Conversely, if a collection event is recorded but the satellite indicates a cloud-free pixel (FLC = 0), the source is classified as dew.

Through this hierarchical integration of ground validation and remote sensing, a filtered and physically consistent time series of mutually exclusive fog and dew events is constructed for the entire study period.

2.3 Thermodynamic Characterization of the Marine Boundary Layer (MBL)

The thermodynamic characterization of the Marine Boundary Layer (MBL) is conditioned by the event-based classification protocol established in Section 2.2. Specifically, timestamps are categorized into fog, dew, or non-events (clear-sky conditions with no atmospheric water collection) based on the criteria recorded at the OYA1211 core station. For each discrete 10-minute interval within these categories, vertical profiles are reconstructed using the altitudinal transect to evaluate structural changes from sea level (56 m a.s.l.) up to the highest operational node (1354 m a.s.l.).

Missing atmospheric pressure data at specific altitudes (z_2) were calculated using the hypsometric equation anchored to the synchronous sea-level barometric observations (P_1) at the coastal airport (z_1) (Campbell and Norman 1998):

$$P_2 = P_1 \exp \left(-\frac{g(z_2 - z_1)}{R \cdot T_{\text{media}}} \right) \quad (2)$$

Where $g = 9.8 \text{ m s}^{-2}$, $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$, and T_{media} (K) is the mean absolute temperature of the air column between both elevations.

2.3.1 Vertical Potential Temperature Gradient

Potential temperature (θ , K) was computed to assess dry adiabatic stability and isolate pressure-induced compression effects along the slope (Zúñiga López and Crespo del Arco 2012):

$$\theta = T \cdot \left(\frac{P_0}{P} \right)^{\frac{R}{C_p}} \quad (3)$$

Where T is the air temperature (K), P is the local pressure (Pa), P_0 is the reference sea-level pressure at the airport (Pa), and $C_p = 1004.67 \text{ J kg}^{-1} \text{ K}^{-1}$. The thermal stratification of the MBL is quantified via the vertical potential temperature gradient ($\Delta\theta/\Delta z$, K m^{-1}):

$$\frac{\Delta\theta}{\Delta z} = \frac{\theta_2 - \theta_1}{z_2 - z_1} \quad (4)$$

Where θ_2 and z_2 correspond to the target meteorological station, and θ_1 and z_1 denote the synchronous reference baseline at the coastal airport.

2.3.2 Vertical Specific Humidity Gradient

Absolute moisture content was derived to track water vapor variations independently of localized thermal fluctuations. In the lower troposphere, specific humidity (q , g kg^{-1}) is physically approximated by the water vapor mixing ratio (r), such that $q \approx r$ (Zúñiga López and Crespo del Arco 2012):

$$q \approx \epsilon \frac{e}{P - e} \quad (5)$$

Where $\epsilon = 0.622$, P is the atmospheric pressure (hPa), and e is the actual vapor pressure (hPa). The term e is extracted from the relative humidity (RH , %) and the saturation vapor pressure (e_s , hPa) derived via Tetens' formulation as a function of temperature (T_c , °C):

$$e = \left[6.11 \cdot 10^{\left(\frac{7.5T_c}{T_c + 237.3} \right)} \right] \cdot \left(\frac{RH}{100} \right) \quad (6)$$

The vertical moisture structure is analyzed through the specific humidity gradient ($\Delta q/\Delta z$, $\text{g kg}^{-1} \text{ m}^{-1}$):

$$\frac{\Delta q}{\Delta z} = \frac{q_2 - q_1}{z_2 - z_1} \quad (7)$$

Where q_2 represents the target station profile and q_1 is the synchronous baseline at the coastal airport.

2.3.3 Vertical Wind Speed Gradient

Mechanical forcing and localized flow modifications induced by the topography are evaluated using the scalar wind speed (WS , m s^{-1}). The vertical wind speed gradient ($\Delta WS/\Delta z$, s^{-1}) isolates shear layers and aerodynamic acceleration along the transect:

$$\frac{\Delta WS}{\Delta z} = \frac{WS_2 - WS_1}{z_2 - z_1} \quad (8)$$

Where WS_2 and z_2 correspond to the target mountain station, and WS_1 and z_1 represent the synchronous wind speed baseline at the coastal airport.

2.4 Temporal Characterization of Fog and Dew Events

To evaluate the evolutionary trajectory and high-resolution dynamics of the Marine Boundary Layer (MBL) during distinct meteorological regimes, a local temporal gradient analysis was performed. For each previously classified timestamp (t_1) identified as fog, dew, or a non-event at the core station (OYA1211), the rates of change for potential temperature (θ), specific humidity (q), and wind speed (WS) were tracked forward in time over a maximum window of 180 minutes.

Temporal variations were computed using finite differences across discrete 30-minute lagged intervals ($\Delta t = 30, 60, 90, 120, 150$, and 180 minutes). The localized temporal gradient for any given thermodynamic property ($\phi \in \{\theta, q, WS\}$) is formulated as follows:

$$\frac{\Delta \phi}{\Delta t} = \frac{\phi_2 - \phi_1}{t_2 - t_1} \quad (9)$$

Where ϕ_1 represents the baseline state of the variable at the initial event detection timestamp (t_1), and ϕ_2 denotes the synchronous value recorded at the subsequent lagged interval (t_2).

This temporal framework tracks how atmospheric conditions evolve step-by-step once an event begins. By analyzing these 30-minute intervals, the methodology captures how quickly air mass properties change (intensify or decay) over time under each environmental regime. Ultimately, these local rates of change help link the data recorded by the weather stations to the broader movement and arrival of maritime air (advective processes) along the mountain slope.

3. Results

3.1 Atmospheric Water Events Characterization and Methodological Comparison

The analysis of the complete annual timeframe for the year 2024 reveals that non-event periods dominated the hyper-arid study site, accounting for 91.41% of the total monitored hours. Consequently, active atmospheric water events, encompassing fog, dew, or co-occurring phenomena, were restricted to a narrow operational window of 8.59% of the overall annual timeframe, as contextualized in the comprehensive baseline framework (Figure 3 A). Despite this shared annual baseline, the internal distribution of these non-rainfall water inputs varied significantly depending on the extraction and classification methodology employed.

Under the Instrument Detection framework (Figure 3 B), which isolates the high-resolution internal structure of the 8.59% active meteorological timeframe, fog and dew dynamics exhibited a highly distributed partition. Pure fog events emerged as the most frequent atmospheric water source, representing 49.2% of the total event duration (4.23% overall). This was closely followed by co-occurring fog and dew events (“Fog + Dew”), which occupied 46.8% of this localized active period (4.02% overall). In contrast, pure dew events were minimal, accounting for a minor 4.0% fraction of the internal instrumentation distribution (0.34% overall).

Conversely, evaluating the same data through the GOES Classification + Visibility methodology (Figure 3 C) forced a shift toward a binary partition due to the structural absence of overlapping or co-occurring classification categories. Under this remote sensing and visibility workflow, pure fog events accounted for 53.3% of the active timeframe (4.58% overall). The remaining 46.7% of the active event timeframe was classified entirely as pure dew events (4.01% overall), effectively absorbing the temporal periods where both atmospheric water mechanisms concurrently overlapped onto the collectors.

To resolve the seasonal variability of these processes throughout 2024, the monthly distribution of mean collected water and relative event frequency was evaluated based on the GOES and visibility classification framework (Figure 4). On a monthly scale, water collection was consistently higher during fog events compared to dew events, revealing a distinct decoupling between yield magnitude and event occurrence rates. This inverse relationship between water volume and event frequency is particularly evident during August, September, and October, a period that simultaneously concentrates the annual peaks of total water collection.

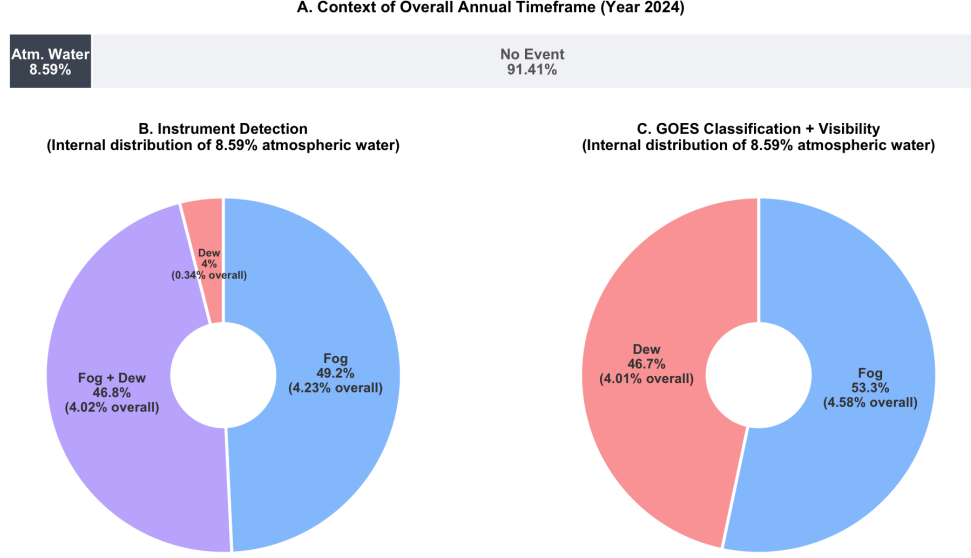


Figure 3: Annual distribution of atmospheric water events during 2024 at the hyper-arid study site. The top horizontal bar (A) establishes the overall annual baseline context, highlighting that active events account for 8.59% of the total timeframe. Inner donut charts display the comparative breakdown within this active window between (B) Instrument Detection and (C) GOES Classification + Visibility methods, showcasing local percentages and their corresponding weight relative to the overall annual timeframe.

During these peak months, mean water yields derived from 10-minute logging intervals reached their maximum annual values. Fog collection peaked sharply in August, reaching nearly 125.0 mL, while dew collection exhibited its maximum accumulation in October, reaching approximately 45.0 mL. In terms of frequency, July represents a unique transitional period where fog and dew occurrences take place in nearly identical proportions. Conversely, the suppressed phase of atmospheric water availability is tightly constrained to June, which exhibited the lowest water collection volumes across both event categories for the entire year.

Complementing the seasonal signal, the mean diurnal cycle of the classified events isolates the sub-daily synchronization of these non-rainfall water inputs (Figure 5). Both fog and dew mechanisms exhibit an asymmetric temporal pattern characterized by maximum water capture during nocturnal hours. During the initial hours of the day, both phenomena display a stable, constant behavior in terms of water yield and occurrence. However, a major structural divergence emerges when cross-referencing frequency and volume metrics. For fog events, higher water collection scales proportionally with higher occurrence frequencies. In contrast,

dew dynamics exhibit an opposite trend, where the periods of lower occurrence frequency synchronize with the highest mean volumes of collected water.

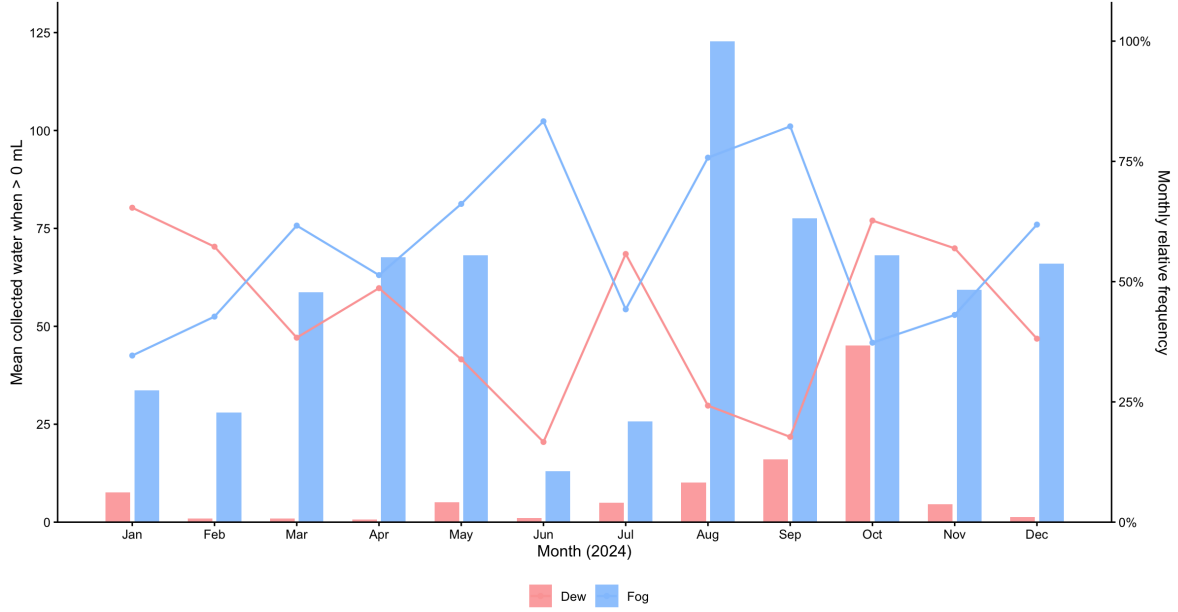


Figure 4: Monthly distribution of classified atmospheric water events during 2024. Bars indicate the mean water collected per event type calculated from 10-minute interval records, while overlaying curves represent the relative frequency of event occurrences for fog (blue) and dew (red).

At the sub-daily scale, the maximum 10-minute interval water yields were recorded during the late afternoon and late night periods. Fog collection achieved its diurnal peak of nearly 300.0 mL at 20:00 UTC, whereas dew collection reached its maximum value of approximately 275.0 mL later in the cycle, at 22:00 UTC. Structurally, a remarkable localized feature occurs at 16:00 UTC, where the relative frequency of dew reaches nearly 100%. This singular peak indicates that out of all the active dew days registered during 2024, the phenomenon was consistently active during this specific hourly window.

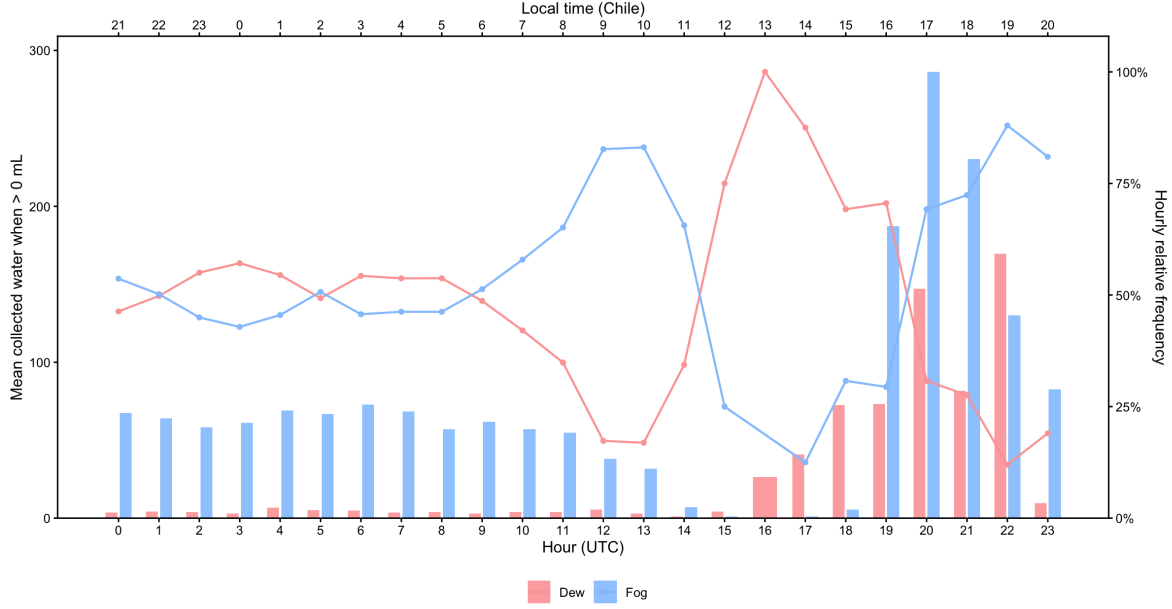


Figure 5: Mean diurnal cycle of classified atmospheric water events during 2024. The sub-daily timeline shows mean water collection volumes from 10-minute registers (bars) and relative hourly frequencies (curves) for fog (blue) and dew (red) events across the UTC timeframe.

3.2 Atmospheric Stability and Boundary Layer Thermodynamics

Following the temporal characterization of the classified events, the mean vertical profiles of specific humidity ($\Delta q/\Delta z$), potential temperature ($\Delta\theta/\Delta z$), and wind speed ($\Delta U/\Delta z$) were evaluated to assess boundary layer stability during the different atmospheric water inputs (Figure 6). To robustly characterize the lower troposphere and boundary layer behavior, these vertical thermodynamic gradients were calculated across the complete altitudinal transect utilizing all eight meteorological stations distributed scaling up Cerro Oyarbide. Because the definitive event classification workflow was anchored strictly at the core node located at 1211 m a.s.l., the corresponding thermodynamic measurements for this specific reference station are highlighted explicitly in red across all parameter distributions to evaluate its local alignment with the regional transect.

For the potential temperature gradient ($\Delta\theta/\Delta z$), the data are contextualized using the thermodynamic thresholds established by Lobos-Roco et al. (2024). In that framework, a baseline threshold of 0.0031 K/m (indicated by the vertical red reference lines) serves to distinguish active atmospheric water events from baseline non-event conditions. When isolating individual event types, the vertical thermal gradients reveal a distinct separation between fog and dew dynamics. Fog events display a narrower distribution centered around a lower threshold of

0.0026 K/m relative to non-event periods. Conversely, dew events shift toward an increased stability signature, clustering around a higher threshold of 0.0034 K/m.

In contrast to the thermal profiles, the mechanical and moisture gradients show a highly homogeneous behavior across all classified categories. For the vertical specific humidity gradient ($\Delta q/\Delta z$), the distribution observed during both fog and dew events remains relatively similar to each other and exhibits no structural departure from the conditions recorded during non-event intervals. Similarly, the vertical wind speed gradient ($\Delta U/\Delta z$) shows overlapping interquartile ranges and no clear statistical differences among the event categories. This statistical homogeneity suggests that while vertical thermal stability acts as a robust indicator across the altitudinal transect, wind shear and specific humidity gradients do not constitute strong physical discriminators to differentiate between fog, dew, or non-event conditions at the study site.

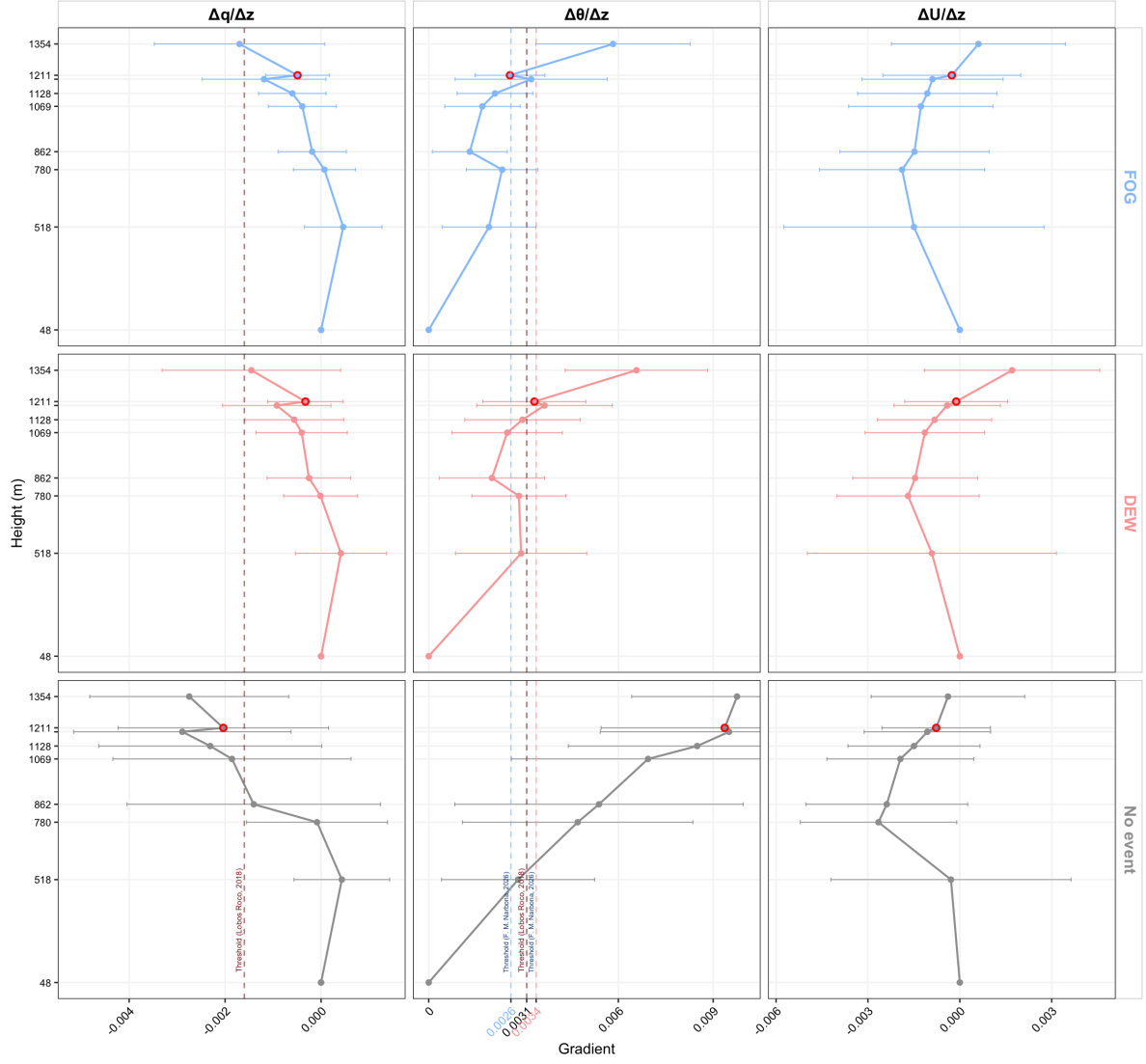


Figure 6: Vertical gradients of boundary layer thermodynamic parameters across classified atmospheric water events during 2024. Boxplots resolve (A) specific humidity gradient ($\Delta q/\Delta z$), (B) potential temperature gradient ($\Delta \theta/\Delta z$), and (C) wind speed gradient ($\Delta U/\Delta z$). Data points corresponding to the classification anchor station at 1211 m a.s.l. are highlighted in red. The horizontal red lines in panel B indicate the diagnostic stability thresholds established by Lobos-Roco et al. (2024).

3.3 Temporal Evolution and Rates of Change

To characterize the immediate atmospheric response during the onset and development of non-rainfall water inputs, the temporal evolution of meteorological anomalies was tracked during the minutes following the initiation of each classified event (Figure 7). For specific humidity anomalies (Δq), fog events display a distinct and progressive increasing trend as the minutes elapse after the event onset. In terms of potential temperature anomalies ($\Delta \theta$), dew events maintain systematically higher values than fog events throughout the entire analyzed temporal window. The wind speed anomalies (ΔU) reveal a consistent decreasing trajectory following the initialization of fog events, a pattern indicative of reduced turbulent mixing once the event establishes at the surface. Conversely, during dew events, wind speed anomalies exhibit no definitive trend, remaining tightly clustered near zero.

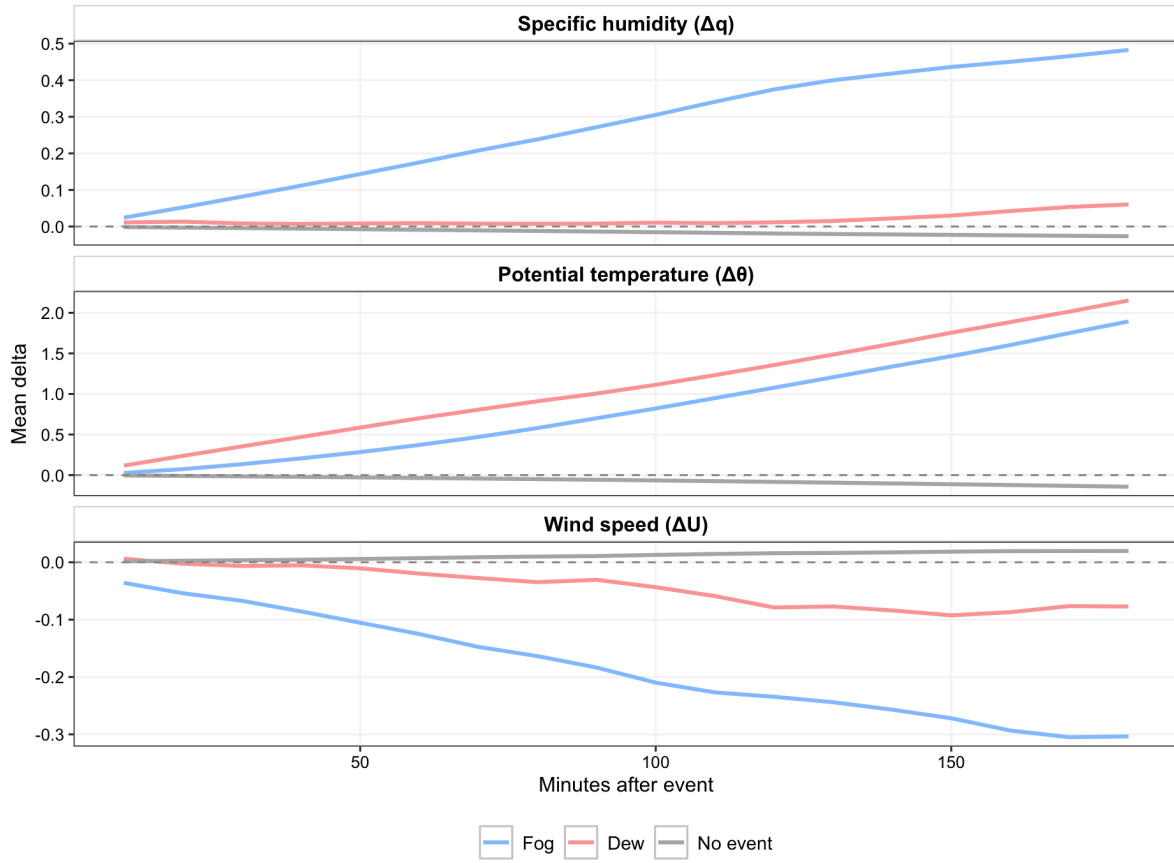


Figure 7: Temporal evolution of boundary layer atmospheric anomalies during the minutes following event initiation for 2024. The lines trace the progressive response of (A) specific humidity (Δq), (B) potential temperature ($\Delta \theta$), and (C) wind speed (ΔU) for fog (blue) and dew (red) events.

To contextualize how these variations align with typical daily cycles, the mean hourly rates of change for each atmospheric parameter were evaluated across a composite 24-hour timeline for the year 2024 (Figure 8). Mirroring the sub-daily distribution documented previously in the diurnal cycle, the highest frequency of dew events concentrates during the afternoon and nocturnal phases. Throughout these specific periods, the temporal rates of change for specific humidity show marked negative values during dew events, capturing a clear depletion of water vapor from the lower atmosphere following event occurrence. In contrast, fog events exhibit significantly smaller fluctuations in their specific humidity rates of change. Regarding potential temperature, dew events display substantially larger rates of change compared to fog across multiple hourly windows, while also exhibiting considerably higher overall variability than both fog and non-event baselines. Conversely, wind speed rates of change display no distinct differences among the event categories. During the early morning hours, the rates of change for all atmospheric parameters converge toward highly similar pathways across all categories, reflecting the minimal process differentiation typical of that phase of the day.

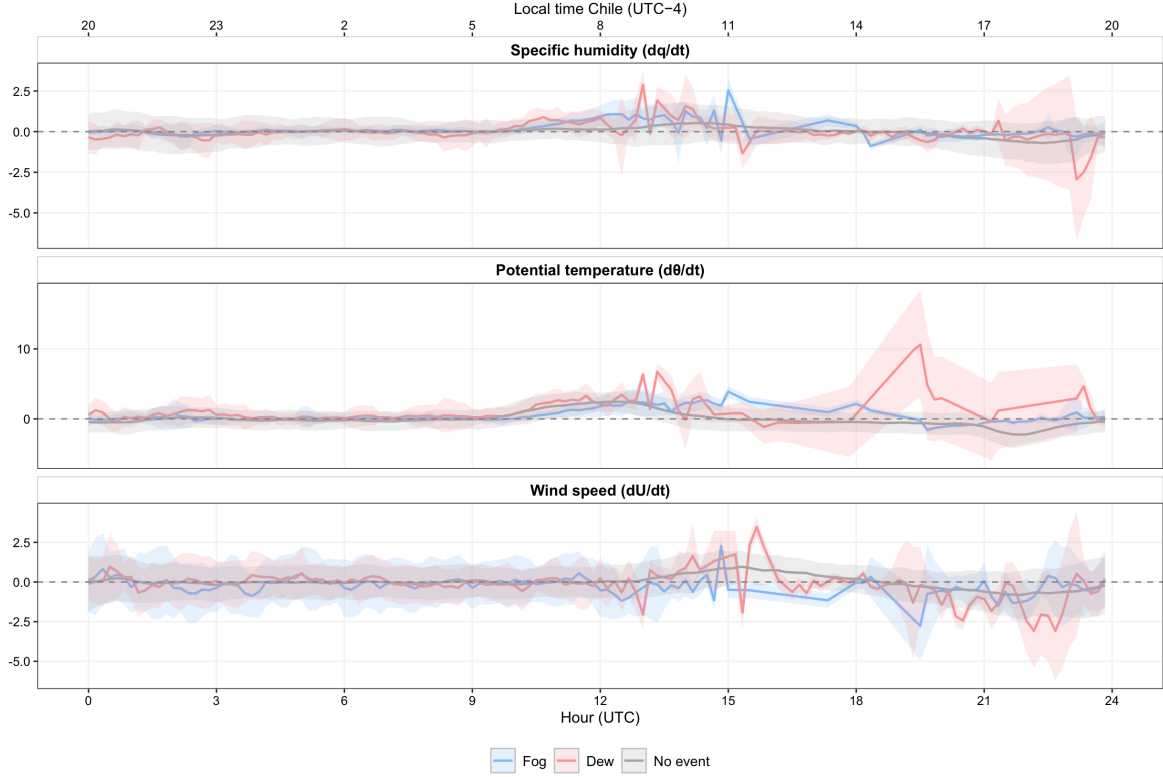


Figure 8: Mean diurnal cycle of temporal rates of change for boundary layer parameters during 2024. The sub-daily timelines display the hourly rates for (A) specific humidity change, (B) potential temperature change, and (C) wind speed change across fog, dew, and non-event intervals.

Finally, the overall statistical distribution and central tendency of these atmospheric rates of change were isolated specifically at the 1-hour post-onset mark to compare overall dataset dispersion (Figure 9). The resulting probability distributions confirm the localized behaviors identified in the preceding timelines. For the specific humidity rate of change (dq/dt), the statistical dispersion and interquartile ranges remain highly comparable between fog and dew events. However, the potential temperature rate of change ($d\theta/dt$) reveals a major structural divergence, where the overall variability during fog events is considerably narrower than that observed during dew events. This heightened dispersion during dew intervals points toward significantly more variable thermal conditions during its development phase. In contrast, the statistical distributions for the wind speed rate of change (dU/dt) overlap almost entirely, indicating highly similar behaviors across fog, dew, and non-event conditions.

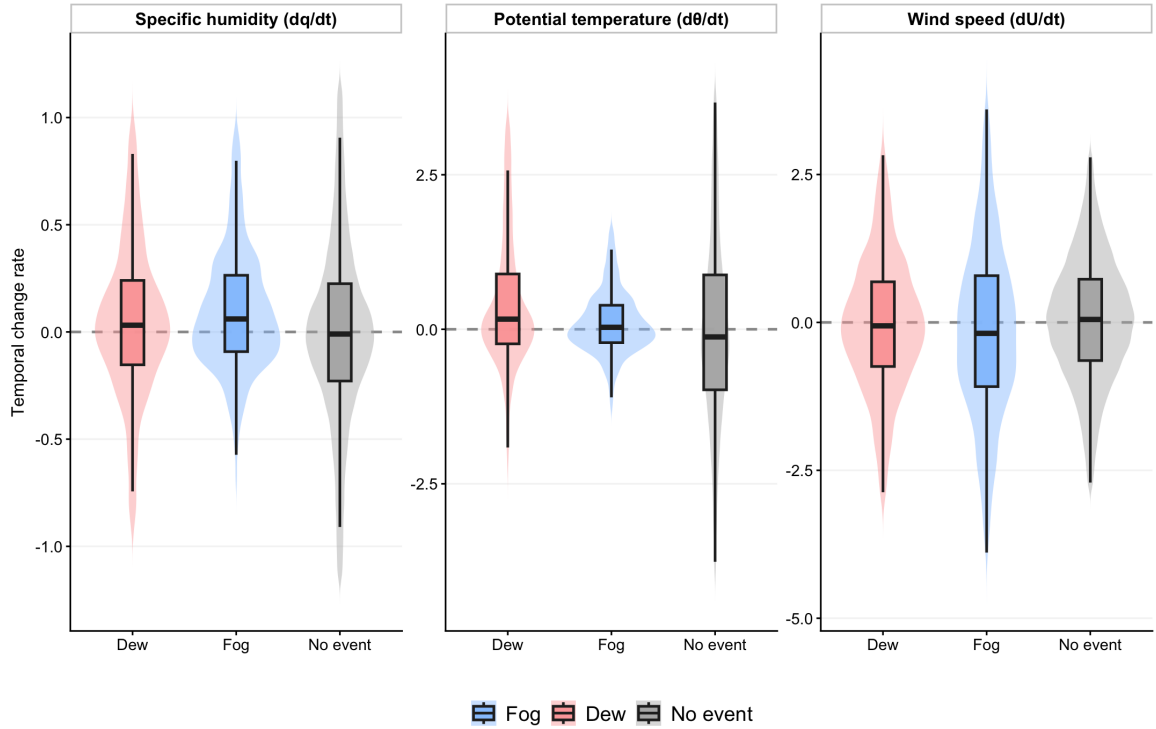


Figure 9: Statistical distribution of boundary layer atmospheric rates of change measured 1 hour after event onset during 2024. The boxplots and density curves resolve the central tendency and dispersion for (A) specific humidity (dq/dt), (B) potential temperature ($d\theta/dt$), and (C) wind speed (dU/dt) under fog, dew, and non-event conditions.

4. Discussion

The comparative analysis between the Instrument Detection and the GOES Classification + Visibility frameworks highlights critical methodological and physical considerations regarding atmospheric water inputs in coastal hyper-arid environments. Although the instrument-based detection indicates a substantial proportion of co-occurring events where fog and dew are collected simultaneously, this architectural configuration is highly susceptible to instrumental crosstalk. Specifically, the passive fog mesh collector can register dew-like inputs via direct water vapor condensation onto its structure, while the dew plate can receive direct droplet deposition during dense fog events.

The application of the definitive GOES-visibility classification framework reveals an extreme quantitative shift, where the relative proportion of pure dew events escalates from a minor 4.0% fraction under the strict instrumental criteria to a substantial 46.7% within the active atmospheric water window. This sharp increase strongly suggests that a major fraction of the events initially interpreted as mixed or co-occurring by ground sensors corresponds predominantly to independent dew deposition. This reclassification reshapes the traditional understanding of hyper-arid hydrometeorology. While surface visibility and GOES satellite imagery are reliable indicators of ground-level fog, both remote and in-situ sensing parameters carry inherent uncertainties. The optical lenses of the visibility sensor can suffer from temporary fogging or condensation artifacts when the instrument cools below the dew point, as structurally documented in the sensor configuration (Figure 2). Furthermore, the GOES satellite accurately resolves low-cloud cover dynamics within the marine boundary layer; however, these stratus decks do not necessarily intersect the complex topography, meaning that cloud detection does not invariably imply effective fog-ground contact.

When assessing these dynamics within a broader regional context, it is critical to note that Cerro Oyarbide represents a highly constrained atmospheric water regime. Active phenomena occupy only 8.59% of the annual timeline, leaving more than 91% of the records as non-events, which inherently limits the total volume of empirical data available for active thermodynamic analysis. This localized hyper-aridity stands in sharp contrast to other prominent coastal fog oases in the Atacama Desert, such as Alto Patache, which exhibit significantly higher annual event frequencies and prolonged cloud-ground intersection windows. Consequently, the decoupled behavior observed on a monthly scale between event occurrence and collection efficiency underscores the highly sensitive thermodynamic thresholds required to trigger deposition at Cerro Oyarbide. Although fog and dew exhibit comparable relative frequencies during specific months, the volumetric yield of fog remains structurally higher, confirming that net accumulation is fundamentally governed by the mechanical intensity and advective efficiency of the marine boundary layer rather than mere occurrence rates. Nevertheless, the fact that dew accounts for 46.7% of the reclassified active window indicates that its ecohydrological role is far more substantial than previously recognized, potentially exerting a vital, continuous impact on the survival of *Tillandsia landbeckii* loma fields during prolonged hyper-arid intervals.

This thermodynamic decoupling is further localized during the sub-daily timeline. The nocturnal phase reveals a stark divergence in the physical mechanisms controlling the diurnal cycle. For dew dynamics, the sharp synchronization between low relative frequencies and elevated water yields implies that dew inputs in this hyper-arid terrain are driven by highly episodic but physically intense condensation windows. This contrasts sharply with fog dynamics, where water accumulation scales proportionally with both volumetric intensity and event recurrence. Collectively, these distinct temporal and diurnal patterns confirm that while fog inputs rely on the continuous advection and structural density of the marine boundary layer, dew accumulation operates under localized, high-efficiency radiative cooling thresholds, establishing two independent non-rainfall water supply mechanisms across the Coastal Cordillera.

The driving forces behind these distinct temporal and diurnal signatures are physically validated by the vertical profiles of the lower atmosphere, showing close alignment with the regional thermodynamic criteria proposed by Lobos-Roco et al. (2024). For the specific humidity gradient ($\Delta q/\Delta z$), the highly homogeneous and overlapping distribution observed between fog, dew, and non-event conditions is entirely expected. Because both fog development and dew formation are fundamentally bounded by near-saturated conditions within the lowest layers of the marine boundary layer, marked variations in vertical moisture stratification are not expected to manifest as a distinct signature between event types.

In contrast, the vertical thermal structure provides a clear differentiation of the atmospheric states. When evaluated against the diagnostic baseline stability threshold of 0.0031 K/m established by Lobos-Roco et al. (2024), the potential temperature gradients ($\Delta\theta/\Delta z$) extracted at the 1211 m a.s.l. core station reveal that dew events cluster significantly above the baseline compared to fog occurrences. This stratification proves that dew events consistently develop under highly stable atmospheric conditions, whereas fog events associate with a relatively more mixed boundary layer state. Such thermodynamic divergence is thoroughly consistent with the underlying physical mechanisms governing both processes; while dew accumulation requires high-efficiency surface radiative cooling under calm, highly stratified boundary layers, fog formation and persistent maintenance are favored by turbulent vertical mixing and continuous cloud-deck advection across the windward slopes of the Coastal Cordillera.

This physical differentiation is further elucidated by the short-term boundary layer adjustment following event initialization (Figure 7). The systematic divergence in specific humidity anomalies (Δq) highlights contrasting moisture source mechanisms. The progressive, linear increase in moisture during fog events is consistent with the continuous advection of humid oceanic air masses, which maintains the near-saturated envelope necessary to sustain fog droplets in suspension. Conversely, the stationary behavior of Δq during dew events confirms that its onset does not rely on advective moisture pulses, but rather on local radiative cooling driving the air toward saturation without modifying the total water vapor mass. In fact, because the phase change involves the direct extraction and deposition of atmospheric water vapor onto cold ground structures, it acts as a localized moisture sink, explaining why specific humidity remains capped or slightly depleted over time. Furthermore, the higher potential temperature anomalies ($\Delta\theta$) coupled with wind anomalies (ΔU) remaining near zero during

dew events solidify the requirement of an undisturbed, highly stratified surface boundary layer. Under these conditions, the suppression of vertical turbulent mixing prevents the destruction of the thermal inversion, whereas the post-onset drop in wind speed during fog events reflects the mechanical adjustment of the boundary layer as the dense, advected stratus deck couples with the local topography.

On a diurnal scale, the structural analysis of the hourly rates of change provides compelling evidence of this differential atmospheric processing (Figure 8). The prominent negative rates of change for specific humidity (dq/dt) during the afternoon and night windows for dew events mathematically capture the surface “drying” effect induced by the condensation process. This localized water vapor depletion confirms the high intensity of the phase change during these sub-daily periods. This behavior stands in contrast to the negligible moisture rates of change observed during fog intervals, where persistent saturation shields the atmosphere from rapid moisture drops. Correspondingly, the accelerated potential temperature rates of change ($d\theta/dt$) and heightened variability during dew intervals confirm that dew dynamics are tightly bound to high-frequency fluctuations in radiative cooling. The stabilization of all parameter rates during the early morning hours, paired with the lack of distinct signatures in wind speed rates (dU/dt), indicates a shared boundary layer state where early-day solar radiation homogenizes the lower troposphere, masking process-specific thresholds.

Ultimately, the statistical boundaries uncoupled at the 1-hour post-onset mark establish the long-term robustness of these atmospheric signatures (Figure 9). The comparable dispersion in dq/dt uncovers that both mechanisms operate at the upper limit of low-tropospheric saturation, requiring identical moisture baselines to trigger initialization. However, the dramatic narrowing of the $d\theta/dt$ probability distribution during fog events confirms that the fog-bound boundary layer operates under highly uniform, structurally mixed thermal conditions, where cloud-deck thickness buffers the surface from intense temperature swings. In contrast, the high dispersion and expanded interquartile ranges during dew events validate the presence of a dynamic, highly sensitive radiative boundary layer governed by local thermal shifts. Finally, the statistical overlap in the dU/dt distributions confirms that wind shear fluctuations function as a background macro-environmental condition rather than a local driving force, validating that atmospheric stability and moisture phase changes are the primary controls governing non-rainfall water inputs across the Coastal Cordillera.

5. Conclusions

This study successfully decoupled the occurrences and boundary layer dynamics of fog and dew events across the complex altitudinal transect of Cerro Oyarbide throughout 2024. By integrating ground-based visibility data and high-resolution GOES-16 satellite observations into a hierarchical workflow, this approach effectively resolved the instrumental crosstalk inherent to standard mechanical collectors. The application of this classification framework revealed an extreme quantitative reallocation of non-rainfall water inputs, where pure dew

events escalated from an unrepresentative 4.0% instrumental fraction to a substantial 46.7% of the active atmospheric water timeframe. This sharp adjustment demonstrates that a major proportion of observations previously labeled as mixed or co-occurring are driven by independent dew deposition, fundamentally reshaping the hydrometeorological baseline of this hyper-arid environment.

Beyond resolving event occurrence, the high-resolution gradient and temporal rate analyses uncoupled the specific atmospheric states and surface couplings that govern each phenomenon. Vertical boundary layer profiles confirmed that while moisture and mechanical wind shear gradients act as homogeneous background conditions across the transect, vertical thermal stability serves as the primary physical discriminator. Validated against regional thermodynamic baselines, fog events systematically align with a lower potential temperature gradient threshold of 0.0026 K/m within a moderately mixed boundary layer, whereas dew events cluster at a higher thermal threshold of 0.0034 K/m, confirming a strict requirement for a highly stratified, stable lower troposphere. This physical differentiation was further validated by the localized sub-daily rates of change. While fog events rely on steady, advective maritime cloud air masses that buffer the surface from intense temperature swings, dew dynamics operate under localized, high-efficiency radiative cooling thresholds. The prominent negative specific humidity rates of change (dq/dt) isolated during peak dew windows mathematically capture a surface “drying” effect, proving that the condensing terrain acts as an active water vapor sink during the nocturnal phase.

In summary, although fog remains the dominant volumetric contributor along the windward slopes of the Coastal Cordillera, the massive representation of dew resolved through this framework underscores its high hidden potential as an independent non-rainfall water resource, indicating that this phenomenon warrants deeper, long-term investigation. This persistent nocturnal condensation may exert a critical, continuous impact on the survival and spatial maintenance of vulnerable *Tillandsia landbeckii* loma fields during prolonged hyper-arid intervals when fog advection is suppressed. Crucially, the entire analytical and computational framework developed in this investigation has been structured to ensure absolute reproducibility. By delivering an open and robust data-processing workflow, this methodology is fully prepared to ingest and analyze future environmental datasets collected at Cerro Oyarbide, establishing a reliable, automated pipeline for long-term boundary layer monitoring and ecohydrological research across the coastal Atacama Desert.

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